

Assignment-1: Histogram

Name: Pennam Nikhil
Roll Number: IMT2024070

[GitHub Repository](#)

Q1. Histograms for id and title

1. Corresponding SQL

PostgreSQL maintains column statistics in the `pg_stats` system view. The following query was used to inspect the histogram statistics for the `id` and `title` columns:

```
SELECT
    attname,
    n_distinct,
    null_frac,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
    AND attname IN ('id', 'title');
```

The table cardinality was verified using:

```
SELECT COUNT(*) FROM title;
```

which returned:

2,528,312

2. What kind of histogram is it?

The PostgreSQL histogram is an **equi-depth histogram**. The histogram boundaries are selected such that approximately equal numbers of sampled tuples occur between consecutive boundaries.

The histogram is generated as part of PostgreSQL's column statistics and is used by the query optimizer for estimating predicate selectivities.

3. How many buckets are there?

The `histogram_bounds` array contains 101 boundary values.

Therefore, the number of histogram buckets is:

$$101 - 1 = \boxed{100}$$

Thus, PostgreSQL uses approximately 100 histogram buckets for both `id` and `title` in the statistics obtained.

4. How is the textual attribute title put into the histogram?

The `title` column contains textual values rather than numerical values. PostgreSQL orders the text values according to its text comparison and collation rules.

The histogram therefore stores textual boundary values in sorted order. For example, boundaries include values such as:

```
'Allo 'Allo!
(#1.140)
(#1.254)
...
(1974-03-04)
...
A Lesson in Murder
Abgeschminkt
...
The Living City
...
Zzaap and the Word Master
```

A text range predicate can therefore be evaluated by determining where the searched value occurs in the ordered sequence of textual boundaries.

The histogram does not require the textual values to be converted into numerical values. Instead, the ordering relationship between strings is used for selectivity estimation.

5. Bucket boundaries

The 101 histogram boundaries obtained for `id` were:

```
114, 24573, 50079, 74448, 102250, 129121, 155133,
181016, 206149, 233863, 260767, 287742, 314506, 339108,
362177, 386919, 411783, 435703, 459964, 485728, 512026,
535879, 560447, 587620, 614036, 640736, 663165, 686493,
710381, 733376, 759225, 782723, 806012, 829052, 852028,
877501, 903058, 927725, 953127, 978610, 1004595, 1032084,
1058076, 1083746, 1108234, 1131865, 1155911, 1179238,
1205508, 1234082, 1259069, 1284674, 1310759, 1335478,
1360463, 1386675, 1412692, 1435435, 1462459, 1488257,
1514089, 1541398, 1568648, 1593315, 1617722, 1642842,
1665716, 1690730, 1717894, 1739710, 1765639, 1790886,
1817383, 1842017, 1864212, 1890074, 1915985, 1943166,
1967914, 1997705, 2023158, 2051349, 2076415, 2103407,
2127836, 2151710, 2175304, 2201304, 2224706, 2251068,
2274191, 2297960, 2324975, 2350972, 2378017, 2401656,
2427498, 2455578, 2477404, 2502294, 2528164
```

The 101 boundaries for `title` were:

```
'Allo 'Allo!, (#1.140), (#1.254), (#1.4825), (#1.80),
(#13.62), (#2.36), (#3.70), (#55.74), (#9.6),
(1974-03-04), (1990-01-27), (1996-07-06), (1999-05-28),
(2001-01-17), (2002-08-20), (2004-03-19), (2005-07-16),
(2006-11-20), (2008-02-07), (2009-09-08), (2011-02-26),
(2012-03-27), (2013-02-22), 30 Years of Emmerdale,
A Lesson in Murder, Abgeschminkt,
Akordet e Kosovës - popullore, An Autopsy of The Killing,
Arrival of Rex, Bad Luck Be a Camper Tonight/Step Clam,
Ben and Burman Find God, Black Bunbustlers, Boy from Woy,
Caffe Lena, Chamana Atha Guntha, Cities of Refuge, Cortège,
Dancing Arabs, Deadly Advice, Der Umweg, Dinty, Dr. Hugo,
Ein Bett für Drei, Empress of Ireland, Extremely Yours, Stryc-9,
Final Approach, Frau Flinz, Genius of the Jet, Goode Lovin',
Hard Times in the Country, Highlights, Hotel Lolita 8,
If Lovin' You Is Wrong, I Don't Wanna Be Rat,
Internet Zombies!, Jed and the Countess, Kadetten,
Kirameite, watashi no naka no watashi, L'ombre d'un doute,
Lagda Ishq Ho Gaya,
Leila Shems: Everyone Is Invited to the Pants Party Except Me,
Little Big Head Man, Love Thy Neighbor,
Männer soll man nicht alleine lassen, Ménage à Trois,
Mob Wives Chicago, Mud Show, Naujawan, Nobody's Darling,
Oh Brother/Taz Babies, Otryv, Path-o-Prasad, Pivete,
Prince William Reveals How the Queen Helped Plan His Wedding,
Rafferty's Rise, Reunions, Royal Cat Nap, Sati Pingala,
Sentenza di morte, Show Biz, Snow White and the Juniper Tree,
Srdele, lignje i leptiri, Summer Solstice Festival, Tap Time,
The Annihilators, The Case of the Horrifying Virus: Part 1,
The Erased, The Haunted Planet, The Living City,
The Nomination of Abraham Lincoln, The Sandman,
The Tramp and the Tenner, This Is My Past, Top 2 Performance,
Turnaround, Une cicatrice derrière la tête, Vertauscht,
Wave Skiing 2.0 Jaws, Who Killed Me?, Y-2 Pappy,
Zzaap and the Word Master
```

6. Histogram bar plots

The PostgreSQL histograms were visualized with value boundaries on the X-axis and frequency on the Y-axis.

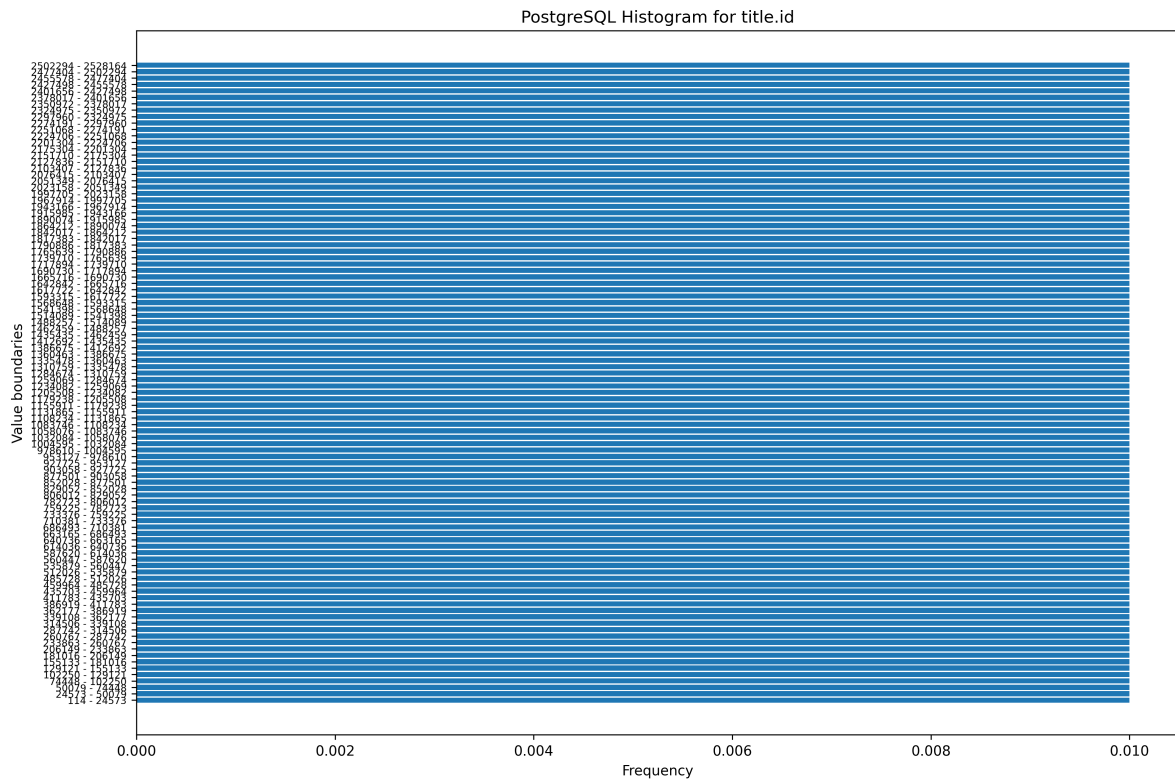


Figure 1: PostgreSQL histogram for the id column

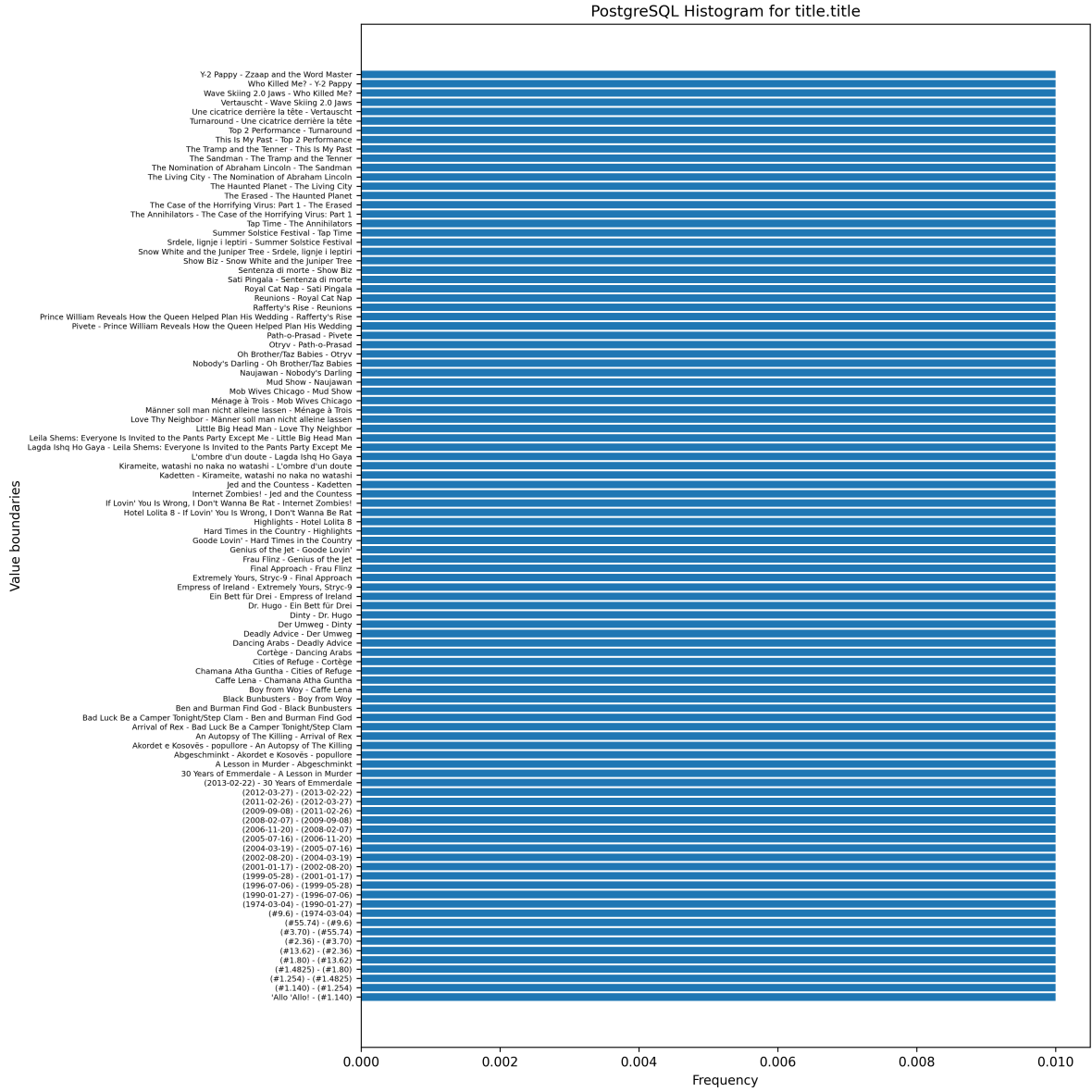


Figure 2: PostgreSQL histogram for the `title` column

Replace the filenames above with the actual Q1 plot filenames.

7. Maximum selectivity error

For each histogram boundary, the estimated selectivity was compared with the actual selectivity obtained directly from the table.

The maximum error is defined as:

$$E_{\max} = \max_b |S_{\text{estimated}}(b) - S_{\text{actual}}(b)|$$

For the `id` histogram, the maximum error occurred at boundary 314506:

$$S_{\text{estimated}} = 0.120000$$

$$S_{\text{actual}} = 0.124393$$

Therefore:

$$E_{\max, id} = |0.120000 - 0.124393|$$

$$E_{\max,id} = 0.004393$$

This corresponds to approximately 0.4393 percentage points.

For the `title` histogram, the maximum error occurred at the boundary (55.74):

$$S_{\text{estimated}} = 0.080000$$

$$S_{\text{actual}} = 0.144477$$

Therefore:

$$E_{\max,title} = |0.080000 - 0.144477|$$

$$E_{\max,title} = 0.064477$$

This corresponds to approximately 6.4477 percentage points.

The larger error for the textual attribute indicates that the distribution of textual values is less accurately represented at some histogram boundaries.

8. Selectivity estimation for a text range query

For a range query on a text attribute, PostgreSQL uses the ordering of textual values and the histogram boundaries.

For example:

```
SELECT *
FROM title
WHERE title < 'Race';
```

The value `'Race'` is located between two histogram boundaries. The optimizer estimates the fraction of tuples below the value based on the position of the value within the histogram.

For a value v between lower boundary L and upper boundary U , linear interpolation can conceptually be represented as:

$$S(v) = S_L + \frac{v - L}{U - L}(S_U - S_L)$$

where the comparison between v , L , and U is based on the ordering of the text values.

This allows PostgreSQL to estimate selectivity even when the predicate is applied to a textual column.

9. Estimated and actual selectivities

The table contains:

$$N = 2,528,312$$

tuples.

For:

```
SELECT * FROM title
WHERE title < 'Race';
```

the query plan estimated 1,917,391 rows and the actual execution returned 1,919,892 rows.

Thus:

$$S_{\text{estimated}} = \frac{1,917,391}{2,528,312} = 0.758368$$

$$S_{\text{actual}} = \frac{1,919,892}{2,528,312} = 0.759357$$

and:

$$|S_{\text{estimated}} - S_{\text{actual}}| = \boxed{0.000989}$$

For:

```
SELECT * FROM title
WHERE id < 350000;
```

the query plan estimated 340,649 rows and the actual execution returned 349,999 rows.

Therefore:

$$S_{\text{estimated}} = \frac{340,649}{2,528,312} = 0.134734$$

$$S_{\text{actual}} = \frac{349,999}{2,528,312} = 0.138432$$

and:

$$|S_{\text{estimated}} - S_{\text{actual}}| = \boxed{0.003698}$$

The results are summarized below.

Table 1: Estimated and Actual Selectivities			
Query	Estimated	Actual	Absolute Error
<code>title < 'Race'</code>	0.758368	0.759357	0.000989
<code>id < 350000</code>	0.134734	0.138432	0.003698

Q2. Serial Equi-Depth Histogram

A Python program was developed to generate and visualize an optimal serial equi-depth histogram. The number of buckets was selected as 20, which lies within the required range of 10–50 buckets.

The implementation connects to PostgreSQL and obtains random samples from the `title` table using the `TABLESAMPLE` command.

The sample sizes used were:

$$\boxed{1000, 3000, 5000}$$

Histograms were constructed independently for the `id` and `title` columns.

1. Histogram bar plots

The required representation uses:

- X-axis: frequency
- Y-axis: value boundaries
- Number of buckets: 20

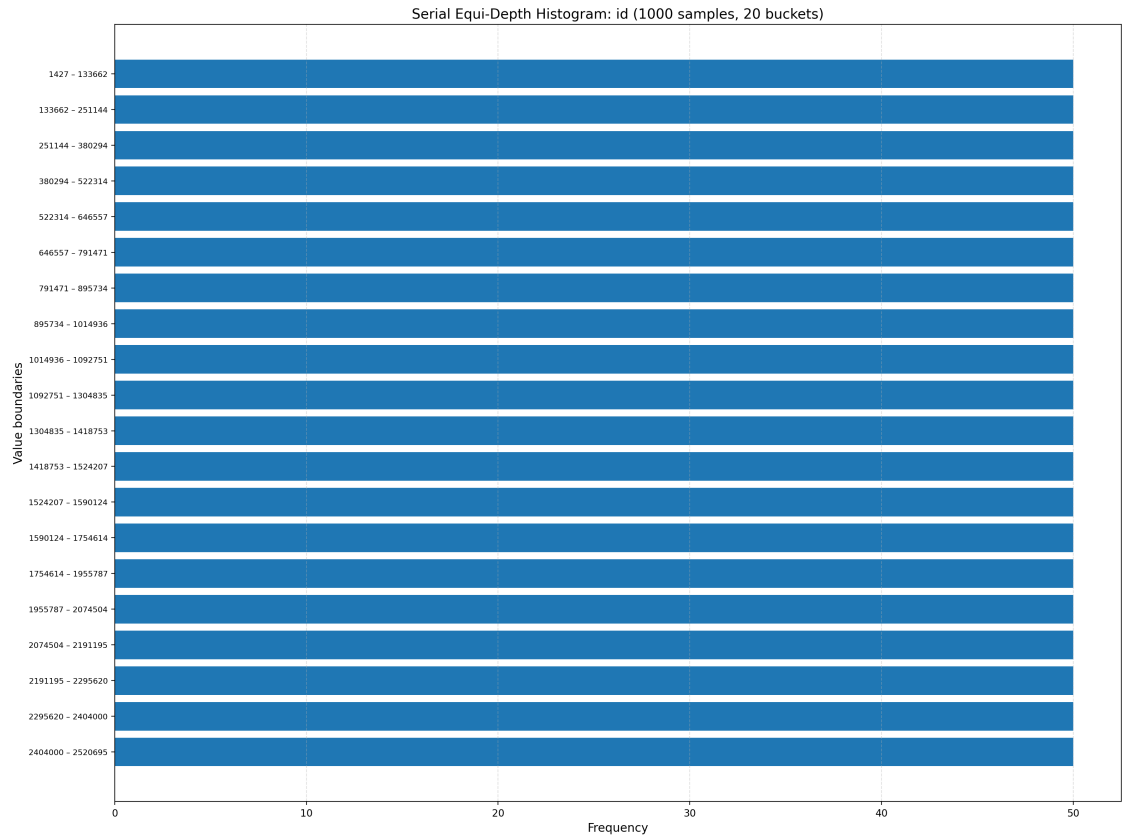


Figure 3: id histogram using 1000 samples

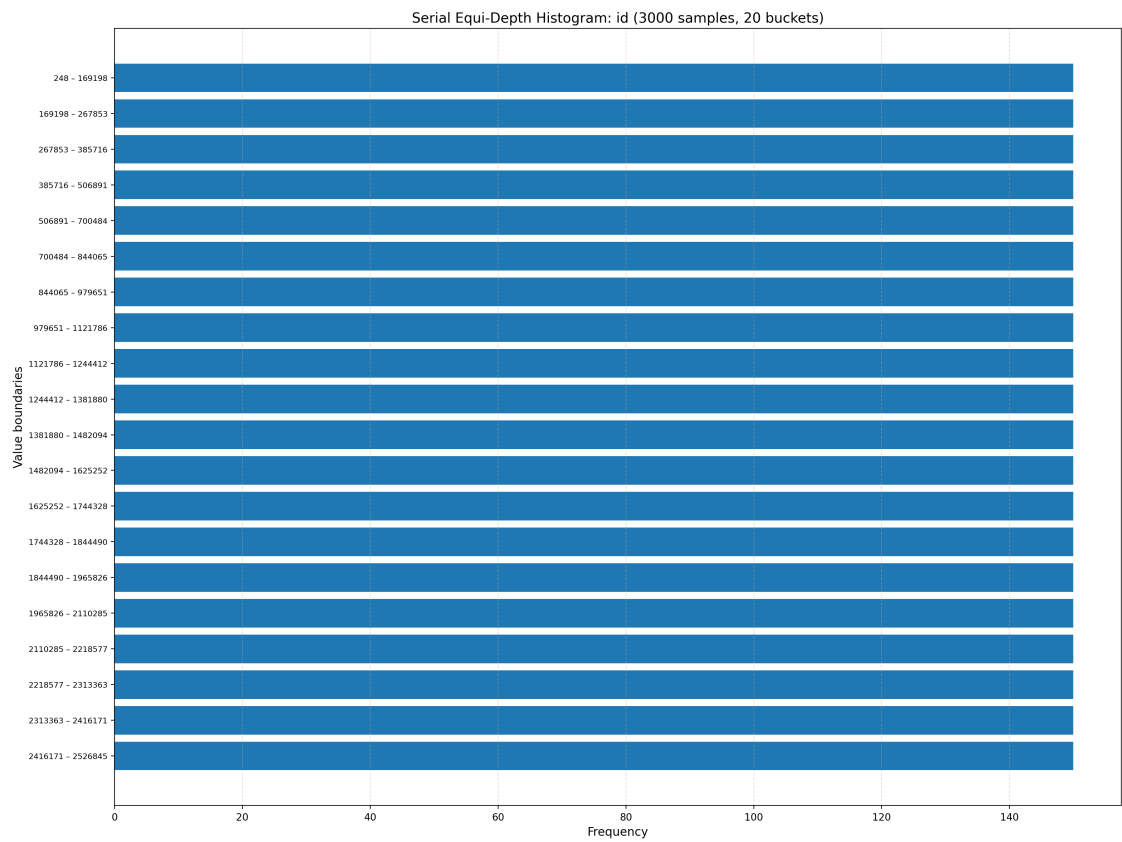


Figure 4: id histogram using 3000 samples

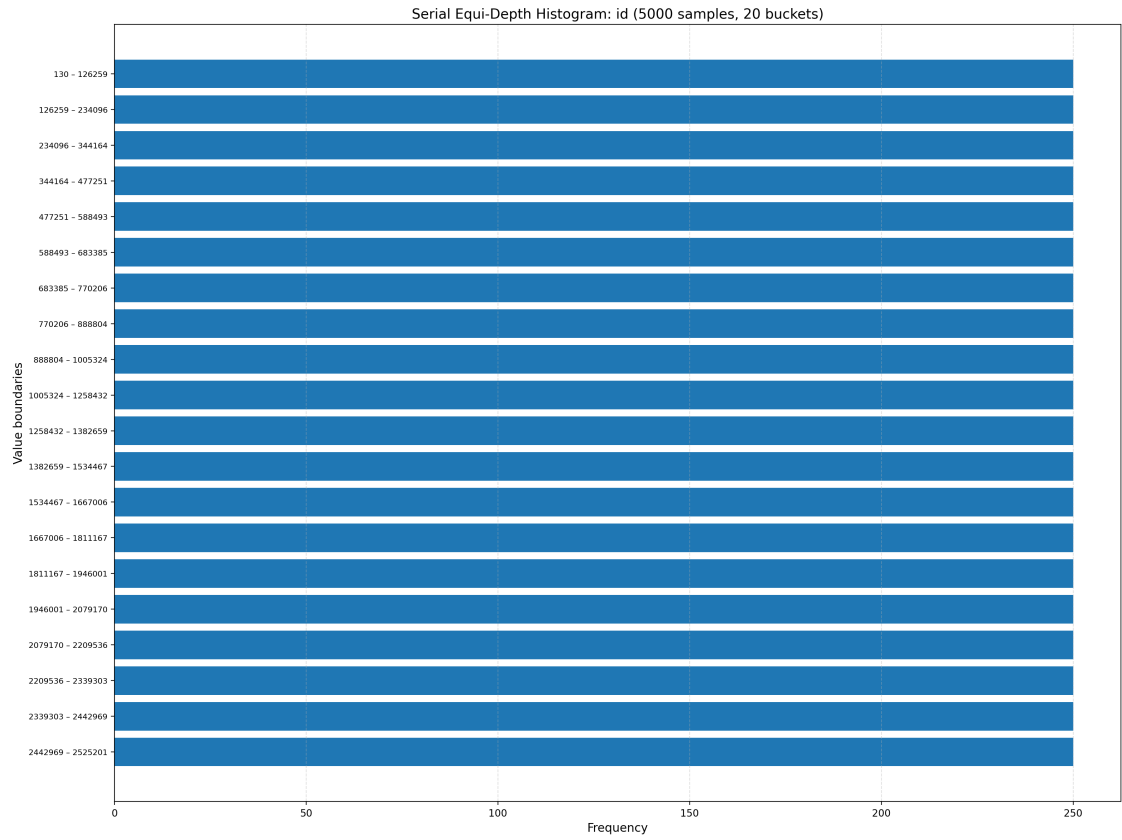


Figure 5: id histogram using 5000 samples

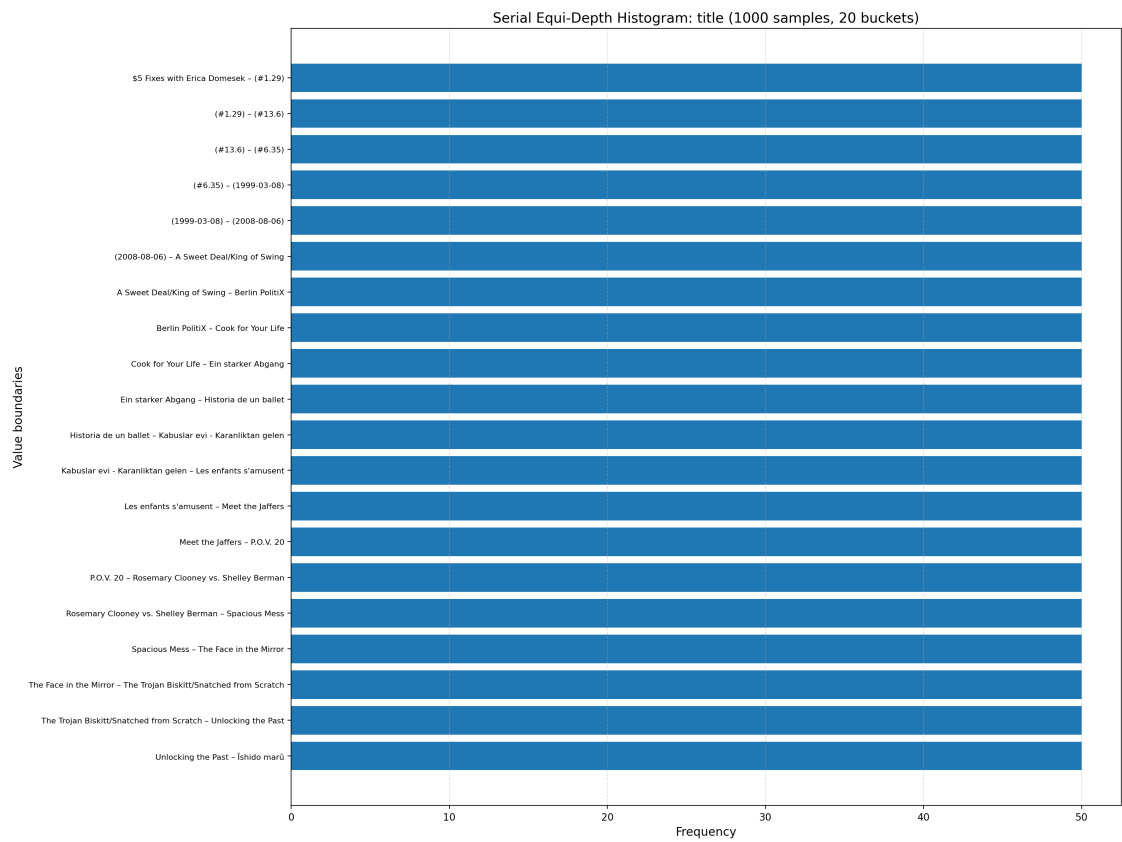


Figure 6: title histogram using 1000 samples

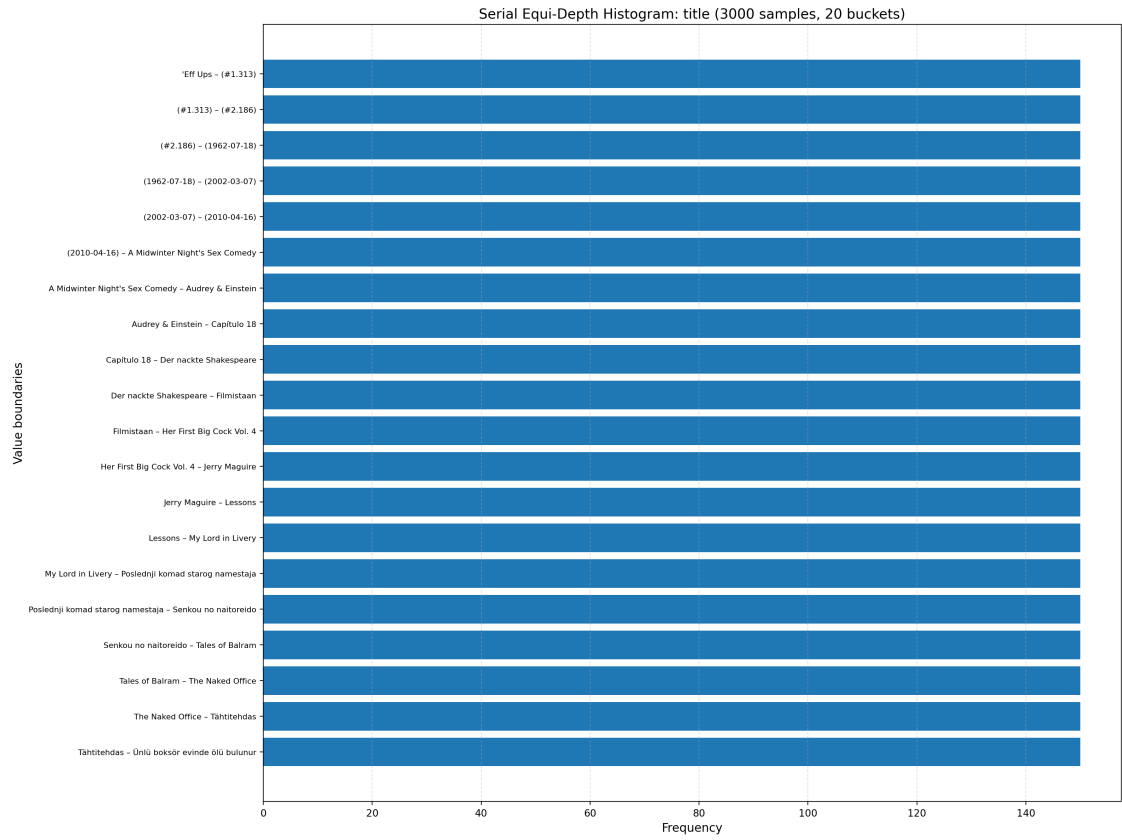


Figure 7: title histogram using 3000 samples

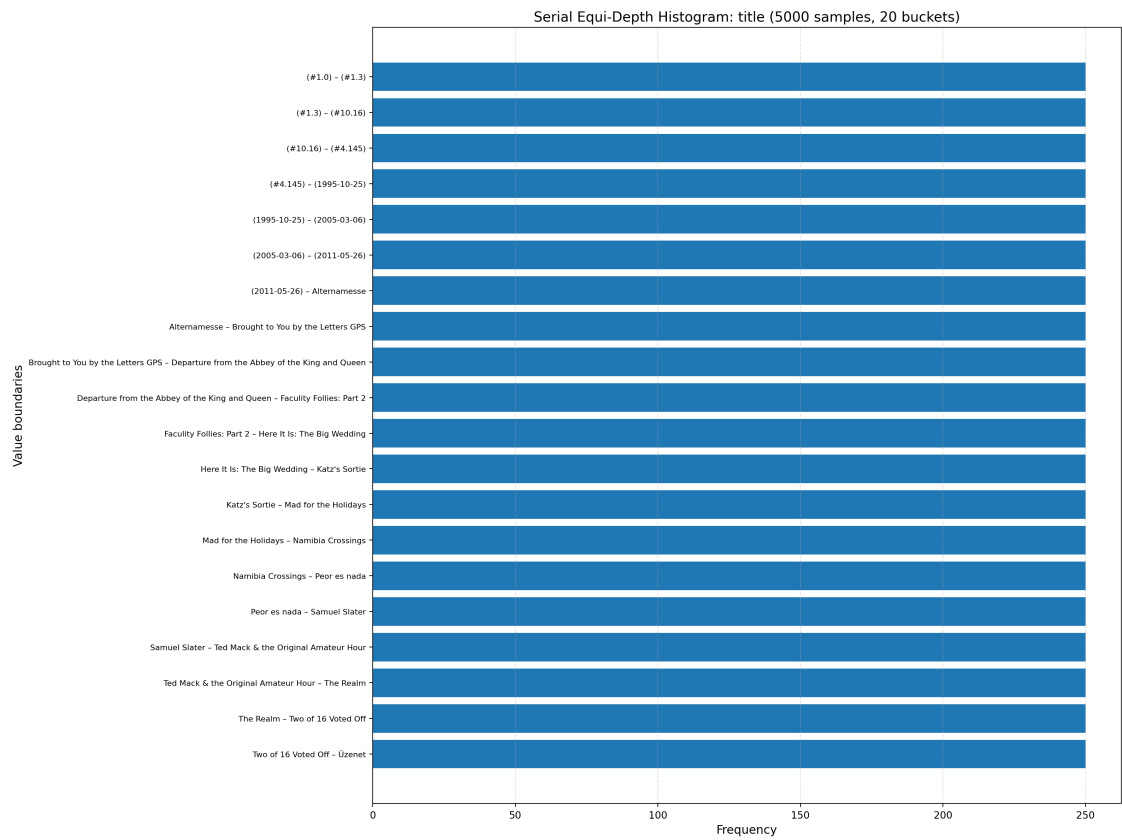


Figure 8: title histogram using 5000 samples

2. Histogram construction time

The measured construction times from the final experiment were:

Table 2: Histogram Construction Time

Sample Size	id Time (s)	title Time (s)
1000	0.000122	0.000251
3000	0.000374	0.000583
5000	0.001368	0.001162

The construction time generally increases as the sample size increases, although the measurements are subject to normal execution and system variation.

3. Frequency bucket boundaries

For 20 buckets and a sample size of 3000, the expected number of tuples per bucket is:

$$\frac{3000}{20} = 150$$

The following boundaries were obtained for the `id` histogram.

Bucket	Lower	Upper	Frequency
1	248	169198	150
2	169198	267853	150
3	267853	385716	150
4	385716	506891	150
5	506891	700484	150
6	700484	844065	150
7	844065	979651	150
8	979651	1121786	150
9	1121786	1244412	150
10	1244412	1381880	150
11	1381880	1482094	150
12	1482094	1625252	150
13	1625252	1744328	150
14	1744328	1844490	150
15	1844490	1965826	150
16	1965826	2110285	150
17	2110285	2218577	150
18	2218577	2313363	150
19	2313363	2416171	150
20	2416171	2526845	150

The `title` histogram boundaries for the same 3000-sample experiment were:

Lower Boundary	Upper Boundary	Frequency
' Eff Ups	(1.313)	150
(1.313)	(2.186)	150
(2.186)	(1962-07-18)	150
(1962-07-18)	(2002-03-07)	150
(2002-03-07)	(2010-04-16)	150
(2010-04-16)	A Midwinter Night's Sex Comedy	150
A Midwinter Night's Sex Comedy	Audrey & Einstein	150
Audrey & Einstein	Capítulo 18	150
Capítulo 18	Der nackte Shakespeare	150
Der nackte Shakespeare	Filmistaan	150
Filmistaan	Her First Big Cock Vol. 4	150
Her First Big Cock Vol. 4	Jerry Maguire	150
Jerry Maguire	Lessons	150
Lessons	My Lord in Livery	150
My Lord in Livery	Poslednji komad starog namestaja	150
Poslednji komad starog namestaja	Senkou no naitoreido	150
Senkou no naitoreido	Tales of Balram	150
Tales of Balram	The Naked Office	150
The Naked Office	Tähtitehdas	150
Tähtitehdas	Ünlü boksör evinde ölü bulunur	150

4. Maximum selectivity error

For each histogram boundary b , the estimated selectivity was compared with the actual selectivity obtained from the full `title` table.

The error is:

$$E(b) = |S_{\text{estimated}}(b) - S_{\text{actual}}(b)|$$

The maximum error is:

$$E_{\max} = \max_b E(b)$$

For the 3000-sample experiment, the measured maximum errors were:

Table 5: Maximum Selectivity Error for 3000 Samples

Column	Maximum Selectivity Error
<code>id</code>	0.045663
<code>title</code>	0.050010

Thus, the maximum selectivity error was approximately 4.57% for `id` and 5.00% for `title`.

5. Repeated sampling for 1000 and 5000 samples

The experiment was repeated using sample sizes of 1000 and 5000, while keeping the number of buckets fixed at 20.

The complete results are:

Table 6: Results for Different Sample Sizes

Column	Samples	Build Time (s)	Max Error
<code>id</code>	1000	0.000122	0.021839
<code>title</code>	1000	0.000251	0.050015
<code>id</code>	3000	0.000374	0.045663
<code>title</code>	3000	0.000583	0.050010
<code>id</code>	5000	0.001368	0.051186
<code>title</code>	5000	0.001162	0.050015

Since TABLESAMPLE produces random samples, the selectivity error does not necessarily decrease monotonically with sample size in a single set of experiments.

6. Extrapolation to the full table

The full `title` table contains:

$$N = 2,528,312$$

tuples.

A linear regression was fitted between sample size and histogram construction time:

$$T(n) = an + b$$

For the `id` column, the fitted model was:

$$T_{id}(n) = 0.000000311250n - 0.000312283461$$

Substituting the full table cardinality gives:

$$T_{id}(2,528,312) \approx \boxed{0.786625 \text{ seconds}}$$

For the `title` column, the fitted model was:

$$T_{title}(n) = 0.000000227625n - 0.000017674950$$

Therefore:

$$T_{title}(2,528,312) \approx \boxed{0.575489 \text{ seconds}}$$

The extrapolated times are summarized below.

Table 7: Extrapolated Full-Table Construction Time

Column	Time (s)	Time (ms)
<code>id</code>	0.786625	786.625
<code>title</code>	0.575489	575.489

These values are extrapolations from the measured histogram construction times and do not represent the time required to scan the entire database.

7. Sample size versus time

The relationship between sample size and histogram construction time is shown below.

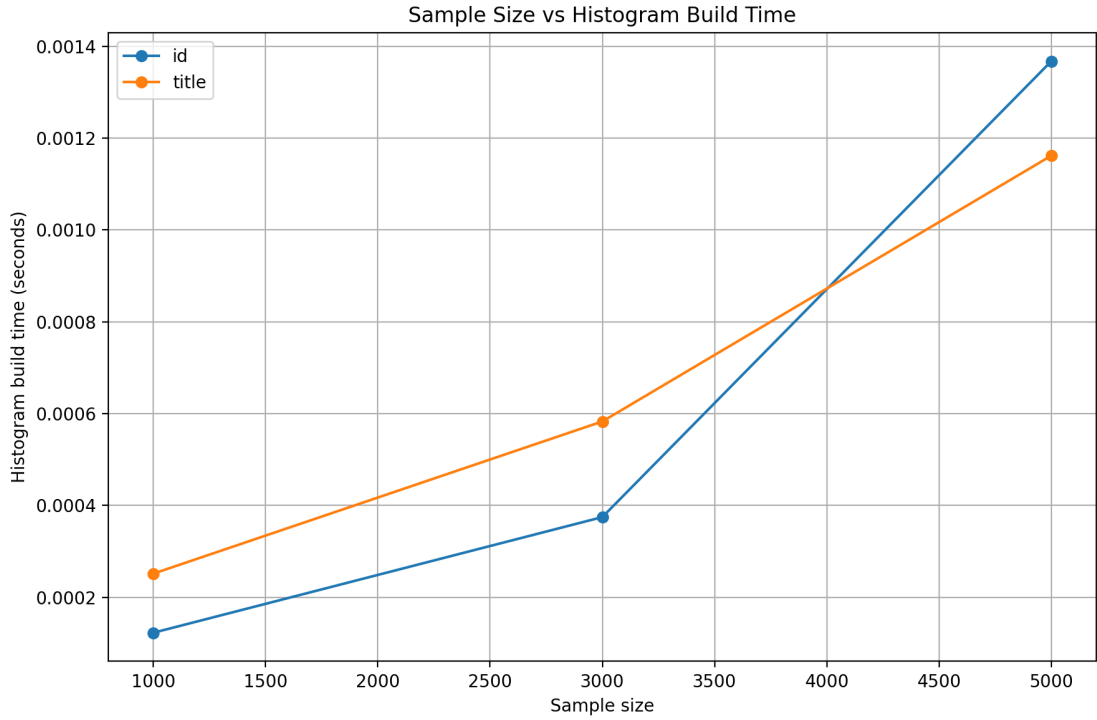


Figure 9: Sample size versus histogram construction time

The overall trend indicates that increasing the number of sampled tuples increases the amount of work required to construct the histogram.

8. Sample size versus maximum selectivity error

The relationship between sample size and maximum selectivity error is shown below.

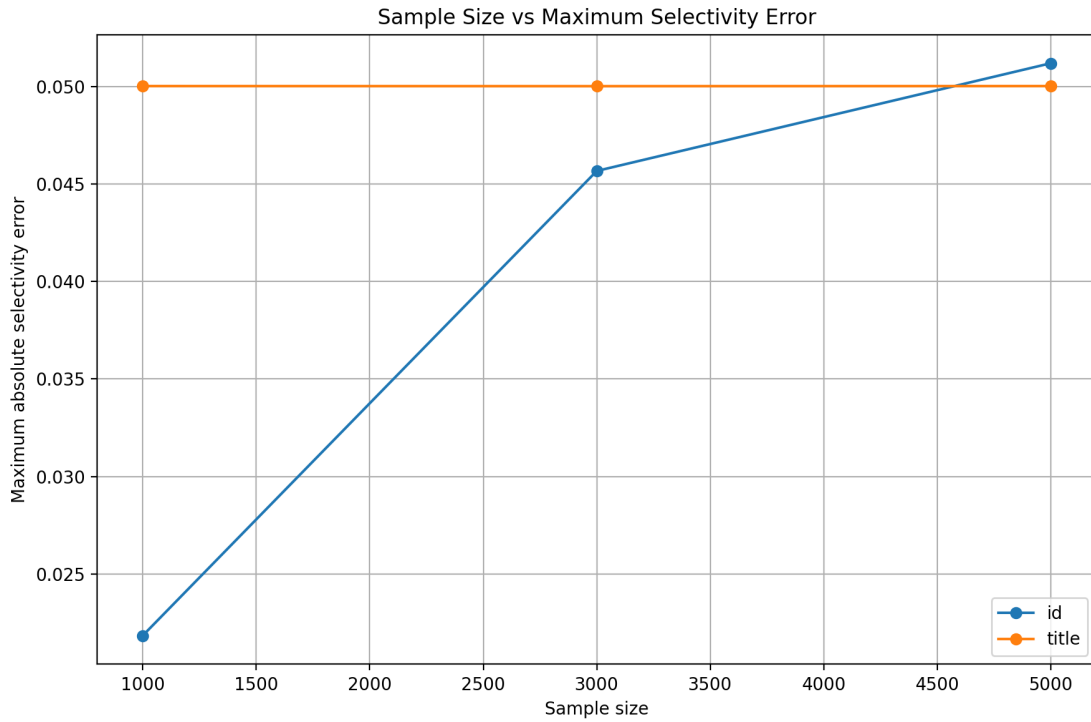


Figure 10: Sample size versus maximum selectivity error

The error is not strictly monotonic because the samples are random. Increasing sample size generally gives more information about the underlying distribution, but individual random samples can produce different errors.

9. Effect of the number of buckets

Using fewer buckets means that each bucket represents a larger range of values. This reduces the amount of information stored in the histogram and can reduce construction overhead. However, the distribution within each bucket is represented less precisely, which can increase selectivity estimation error.

Using more buckets provides a finer representation of the data distribution. This can improve selectivity estimation because each bucket covers a smaller range. However, more buckets require additional memory and processing time.

Therefore, there is a trade-off:

Fewer buckets → lower cost but lower resolution

More buckets → higher cost but higher resolution

The experiment uses 20 buckets as a balance between these two factors.

Conclusion

This assignment examined both PostgreSQL's built-in histogram statistics and a custom serial equi-depth histogram implementation.

For Q1, PostgreSQL's statistics for the `id` and `title` columns contained 101 histogram boundaries, corresponding to 100 buckets. The maximum selectivity errors calculated from the histogram boundaries were 0.004393 for `id` and 0.064477 for `title`.

For the two specified range queries, the selectivity error was 0.000989 for `title < 'Race'` and 0.003698 for `id < 350000`.

For Q2, a custom serial equi-depth histogram with 20 buckets was implemented using random samples obtained with PostgreSQL's `TABLESAMPLE` command. Experiments were performed using 1000, 3000, and 5000 samples for

both `id` and `title`.

The results demonstrate the trade-off between sample size, histogram construction time, number of buckets, and selectivity estimation accuracy. A larger number of buckets provides greater resolution but requires additional processing and storage, while fewer buckets reduce the cost at the expense of estimation precision.